

# **Interference Detection and Classification in Open RAN Using Neural Networks**

ECE 542 – Group 9  
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# Motivation and Problem Statement

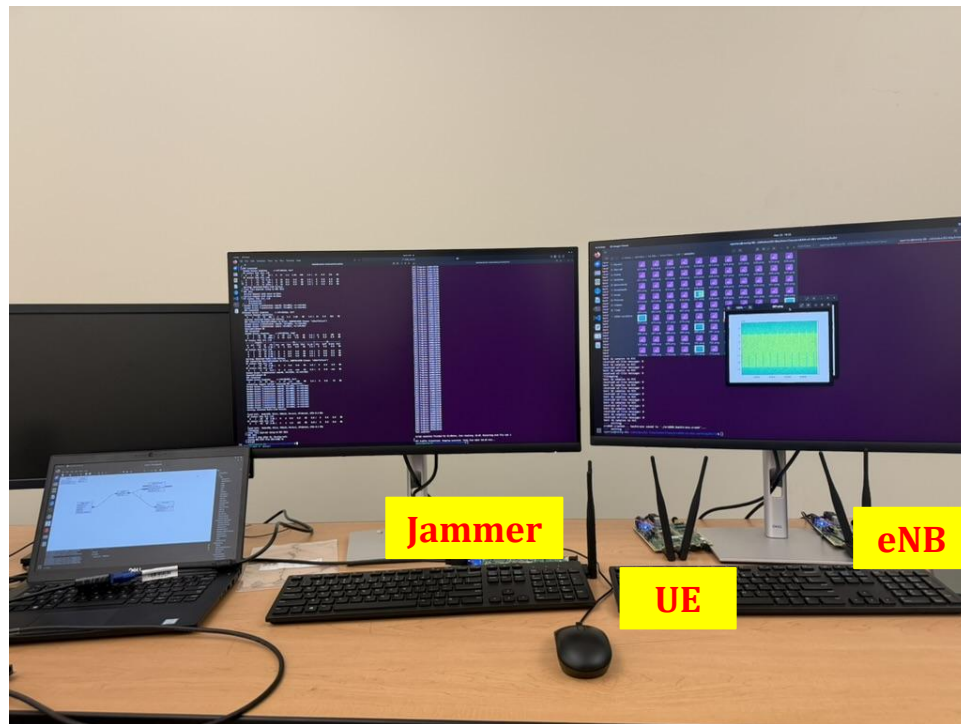
## → Motivation

- ◆ In a typical wireless communication network such as O-RAN, real-time interference detection is key to ensure optimal network performance and QoS. Various interference forms can severely degrade the LTE network, requiring fast and accurate detection and control.

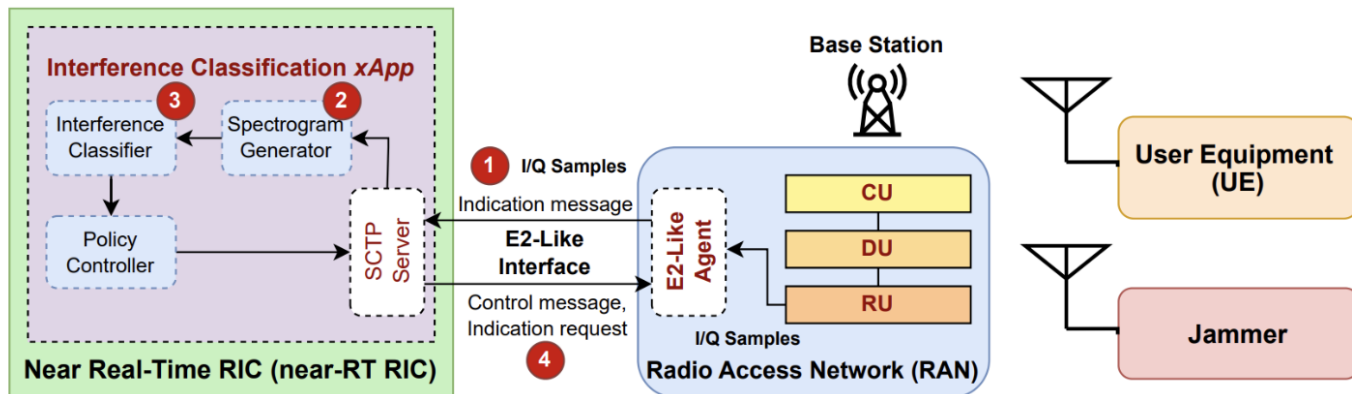
## → Problem Statement

- ◆ Develop a neural network-based **xApp** that detects and classifies interference (SOI / CWI / CI) **within 10 ms – 1s**, matching near-RT RIC control loop timing.
- ◆ The challenge is balancing **classification accuracy** and **inference speed** under strict real-time latency constraints.

# Dataset Source: LTE O-RAN Testbed



# High-level Overview of the Closed-Loop Control Operation



# Inputs and Outputs Descriptions

## → Input: Spectrogram Images (128 x 128 x 3)

- ◆ Visual representation of the channel
- ◆ SOI - No interference
- ◆ SOI+CWI - Continuous wave interference
- ◆ SOI + CI - Chirped interference

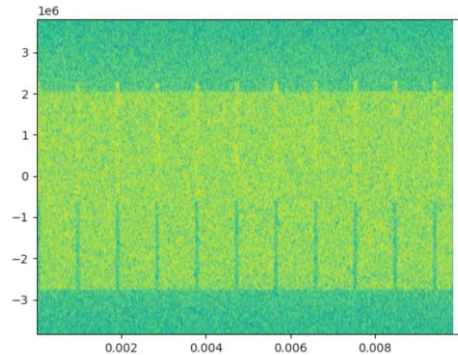
## → Output

- ◆ Predicted Class: {SOI, SOI+CWI, SOI+CI}
- ◆ Interpreted by the xApp to trigger appropriate interference mitigation mechanisms - switch to adaptive modulation scheme

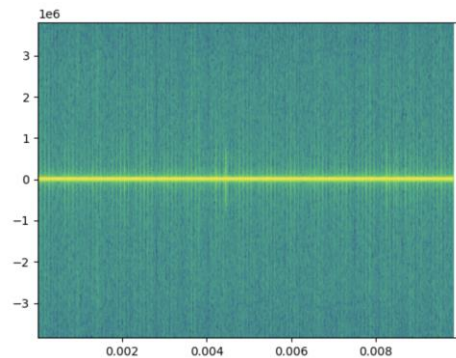
# Exploratory Data Analysis

# Dataset Overview

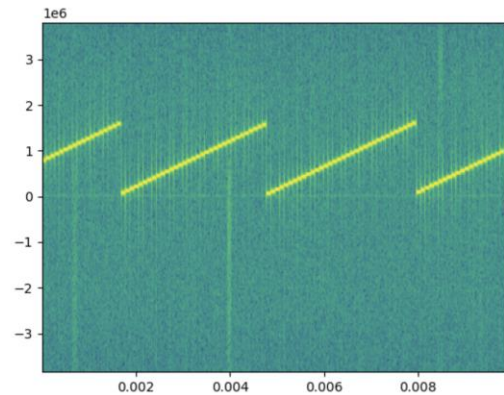
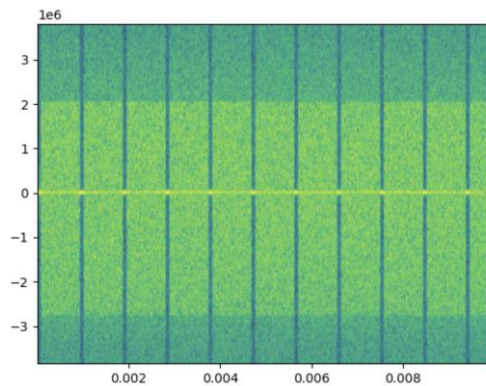
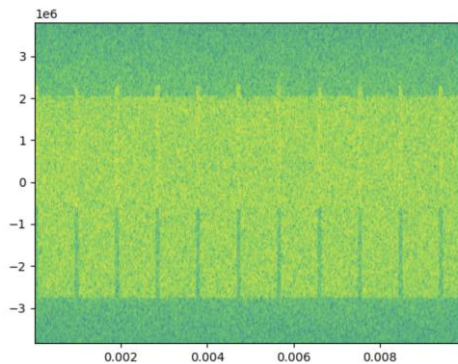
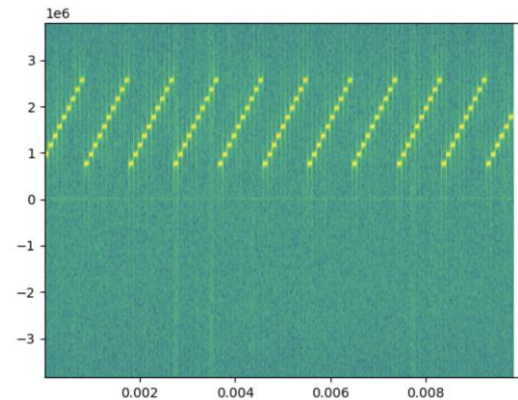
SOI

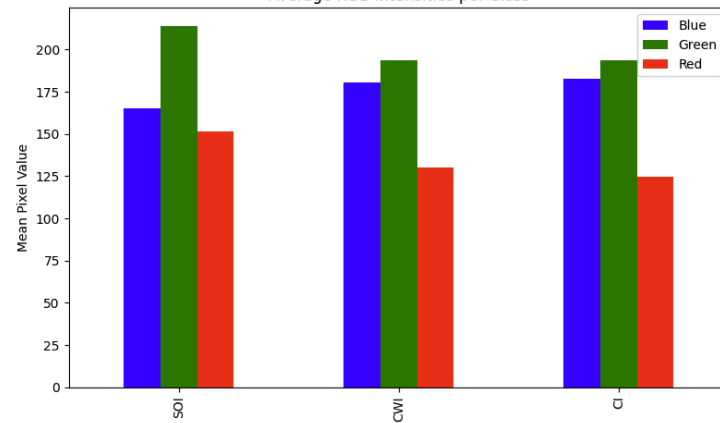
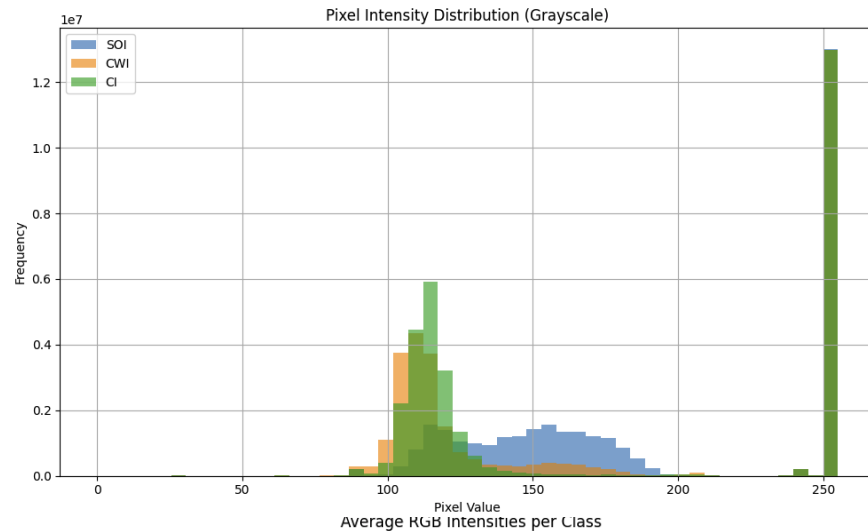
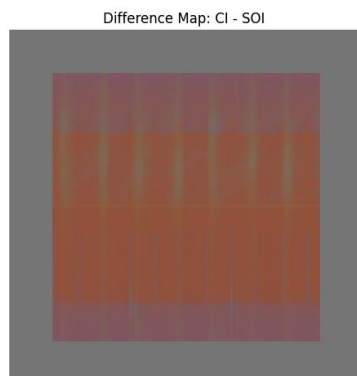
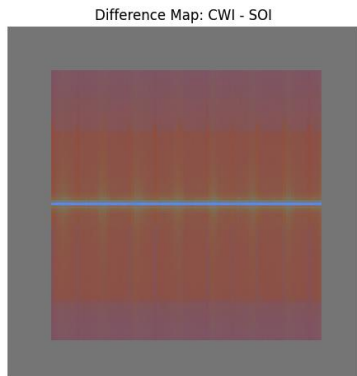
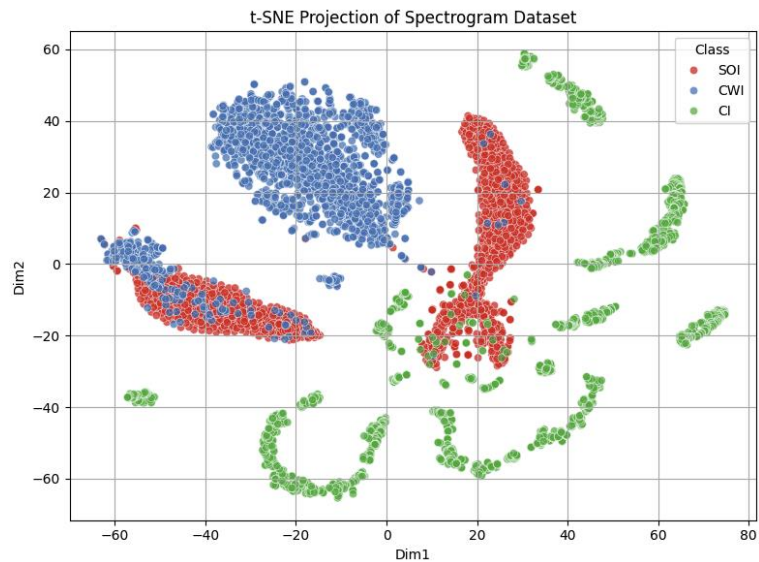


CWI



CI







# **Methodology and Models Description**

## → Data Preparation:

- ◆ Dataset split: 70% training (4200 images), 20% validation (1200 images), 10% test (600 images).
- ◆ Images scaled to [0,1] and flattened for model input.
- ◆ The spectrogram images are resized to (256 x 256 x 3) for MLP and (224 x 224 x 3) for EfficientNet
- ◆ Data Augmentation[EfficientNet ONLY]: random horizontal flip and brightness adjustment

## → Model Training

### ◆ MLP:

- Optimizer: Adam (learning rate = 0.0001).
- Loss: Sparse categorical cross-entropy.
- Trained for 10 epochs on Google Colab with A100 GPU

### EfficientNet:

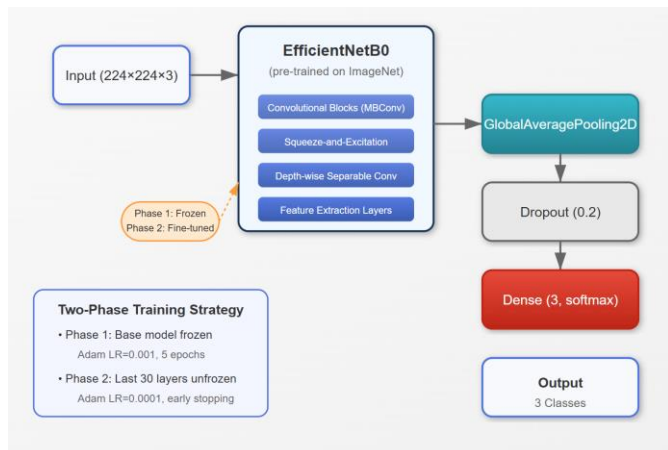
- Phase 1: Base model frozen, Adam LR=0.001, 5 epochs
- Phase 2: Last 30 layers unfrozen, Adam LR=0.0001, early stopping

## → Metrics Used:

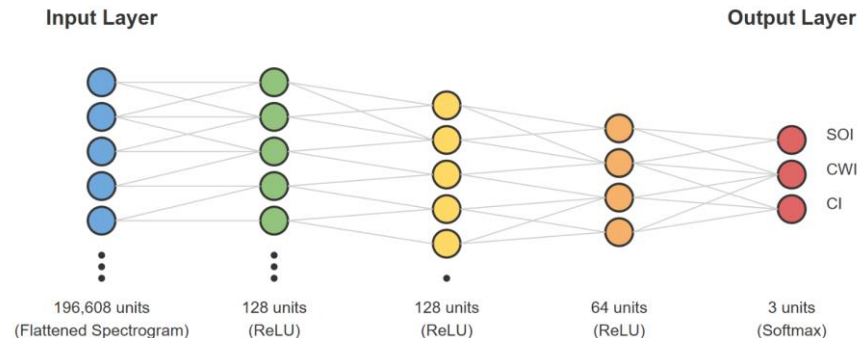
- ◆ **Accuracy:** Proportion of correctly classified test samples.
- ◆ **Precision:**  $\text{True positive} / (\text{True positives} + \text{False positives})$
- ◆ **Recall:**  $\text{True positives} / (\text{True positives} + \text{False negatives})$
- ◆ **F1 Score:** Harmonic mean of precision and recall
- ◆ **Confusion Matrix:** Visualize true vs. predicted labels.

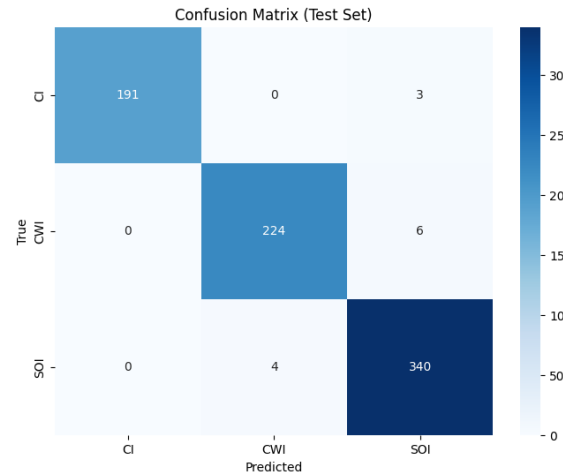
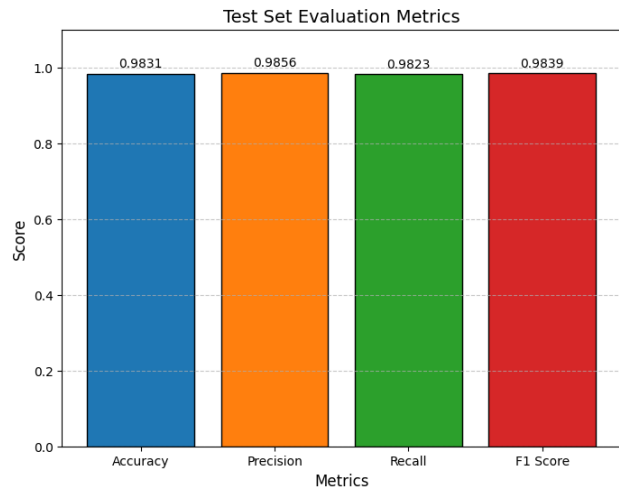
**Model Architecture:**

- Input Layer: RGB spectrogram (150,528).
- Bae Model: EfficientNetB0 (pre-trained on ImageNet).
- Custom Layers:
  - ◆ GlobalAveragePooling2D
  - ◆ Dropout
  - ◆ Dense: 3 units (softmax)

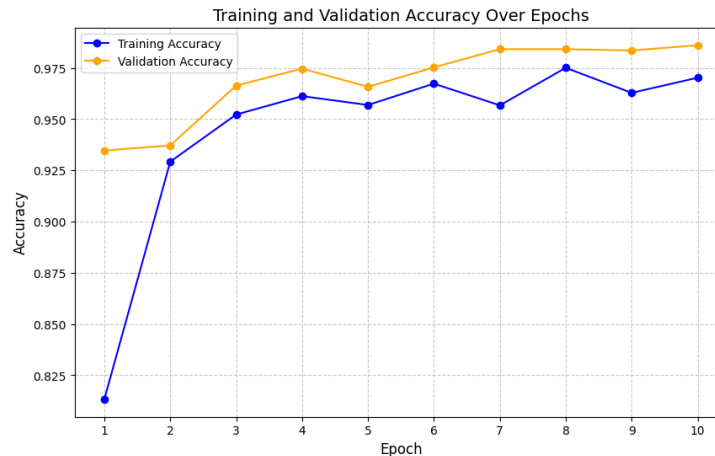
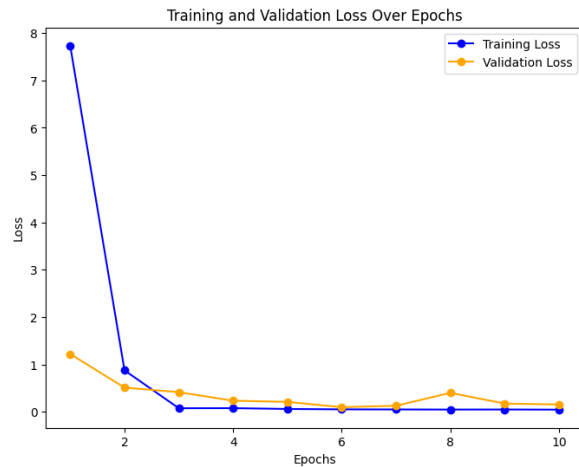
**MLP Model Architecture:**

- Input Layer: Flattened spectrogram (196,608).
- Hidden Layers: 128 units (ReLU), 128 units (ReLU), 64 units (ReLU).
- Output Layer: 3 units (softmax) for SOI, CWI, CI.

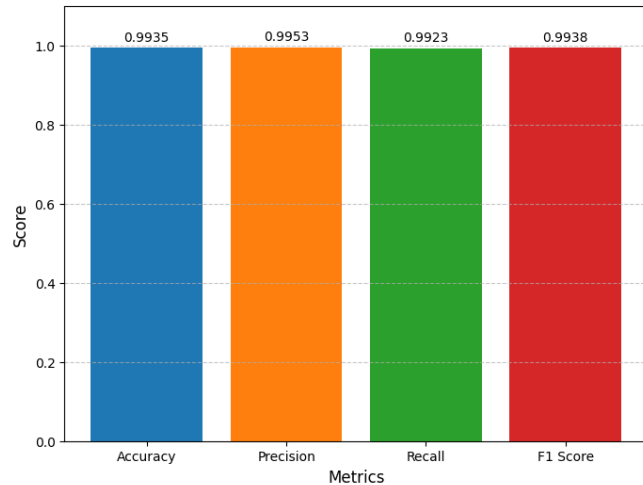




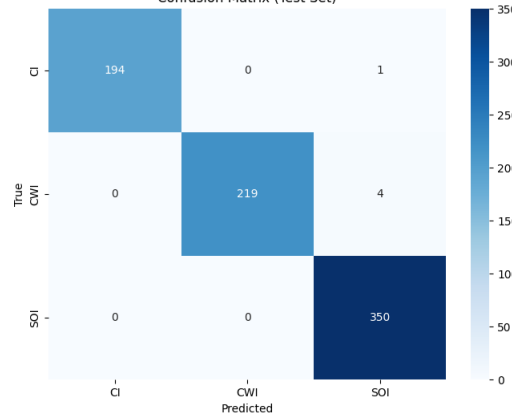
**Average Inference  
Time - 2 milliseconds**



Test Set Evaluation Metrics

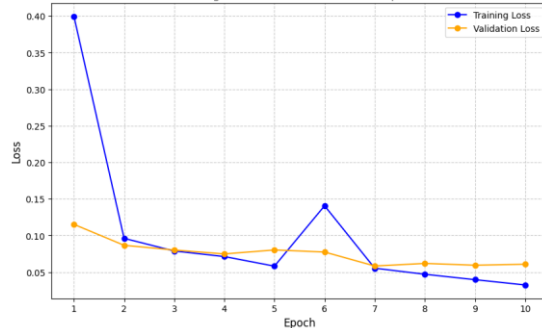


Confusion Matrix (Test Set)

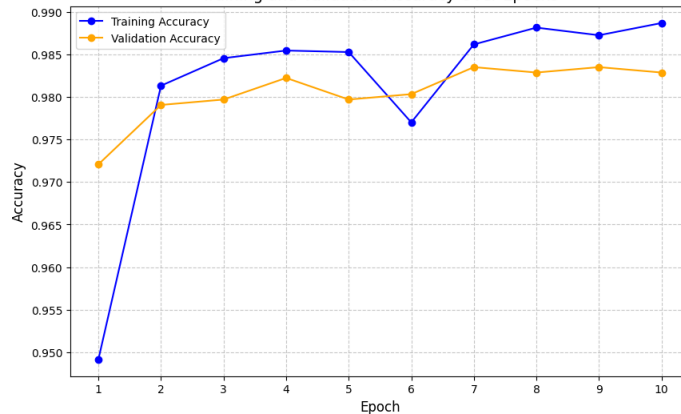


**Average Inference  
Time - 8.96  
milliseconds**

Training and Validation Loss Over Epochs



Training and Validation Accuracy Over Epochs



# Prior Work

- Hands-on experience with LTE O-RAN testbed and xApp development
- Experience with machine learning

# References

- Stephenson, Nathan H., Azuka J. Chiejina, Nathaniel B. Kabigting, and Vijay K. Shah. "Demonstration of closed loop ai-driven ran controllers using o-ran sdr testbed." In *MILCOM 2023-2023 IEEE Military Communications Conference (MILCOM)*, pp. 241-242. IEEE, 2023.
- Chiejina, A., Kim, B., Chowhdury, K. and Shah, V.K., 2024, May. System-level analysis of adversarial attacks and defenses on intelligence in O-RAN based cellular networks. In *Proceedings of the 17th ACM Conference on Security and Privacy in Wireless and Mobile Networks* (pp. 237-247).



**Thank You**